# Repcard CRM Integration

## 1. Overview

### 1.1 Description

**Repcard CRM Integration** connects **Repcard CRM** (field sales CRM for solar, home services, and insurance) with **Newo Platform**.

It enables:

- Creating appointments in Repcard with customer details, address, and calendar selection
- Cancelling existing appointments by updating their status
- Checking availability by computing free 30-minute slots from existing appointments
- Creating contacts (customers) in Repcard with type, owner, and status assignment
- Sending digital business cards (RepCards) to contacts via SMS or email
- Dynamic calendar synchronization from the Repcard API at setup
- DevMode for testing without real API credentials (LLM-emulated responses)
- Auto-configuring conversation canvas with booking and cancellation scenarios

This integration is designed for **field sales teams** (solar installers, home services providers, insurance agents) who use Repcard CRM to manage their leads and appointments and need **an AI receptionist to book, cancel, and check availability** through voice or chat.

### 1.2 Use Cases

- When a caller requests an appointment --> Agent collects details (name, phone, email, address, date/time) and creates an appointment in Repcard
- When a caller wants to cancel an appointment --> Agent finds the booking and cancels it in Repcard
- When a caller asks about availability --> Agent fetches existing appointments and computes free 30-minute slots over the next 5 days
- When a contact needs to be created --> Agent collects customer info and creates a contact in Repcard via POST /customers
- When a digital business card needs to be sent --> Agent sends a RepCard to a contact via SMS or email
- When the project is published --> Agent syncs available calendars from Repcard API and populates the calendar dropdown

### 1.3 Supported Features

| Feature | Supported | Notes |
|---|---|---|
| Create Appointment | Yes | Full customer details, address, calendar selection, setter email |
| Cancel Appointment | Yes | `PATCH /appointments/{id}` with cancellation notes |
| Check Availability | Yes | Computes free 30-min slots from existing appointments (5-day window) |
| Create Contact | Yes | POST /customers with type, owner, status |
| Send RepCard | Yes | POST /send-repcard (digital business card via SMS/email) |

| | | |
|---|---|---|
| Calendar Sync | Yes | Dynamic sync from GET /calendar/lists on publish |
| Canvas Auto-Setup | Yes | Booking + cancellation scenarios and intents |
| DevMode (LLM Emulation) | Yes | Test without real API -- LLM generates realistic responses |
| Reschedule | No | Cancel + rebook as workaround |
| Existing Client Lookup | No | Not implemented |

## 2. Requirements

Before installation:

- Active **Repcard CRM** Enterprise account with API access
- **API Key** from Repcard Company Settings > API Keys tab
- At least one calendar configured in Repcard
- A setter (sales rep) email address for appointment assignment
- A contact owner email for contact creation

## 3. How to Setup

### 3.1 Step 1 -- Get API Credentials from Repcard CRM

1. Log in to your **Repcard CRM** account
2. Navigate to **Company Settings** > **API Keys** tab
3. Generate or copy your **API Key**
4. Note: API access requires an **Enterprise** account
5. Note your **Calendar IDs** (these will be synced automatically on publish)
6. Note the **setter email** address for appointment assignment

### 3.2 Step 2 -- Connect in Platform

1. Open **Newo --> Projects**
2. Set the following attributes in **Builder / Attributes**:

| Attribute | Required | Default | Description |
|---|---|---|---|
| `repcard_api_key` | Yes | -- | API Key from Repcard Company Settings > API Keys tab |
| `repcard_base_url` | No | `https://api.repcard.com` | API base URL |
| `repcard_mode` | No | `Production` | `Production` for real API, `DevMode` for LLM emulation |
| `repcard_enable_booking` | No | `True` | Enable appointment creation |
| `repcard_enable_cancellation` | No | `True` | Enable appointment cancellation |

| | | | |
|---|---|---|---|
| `repcard_enable_contact_creation` | No | `True` | Enable contact creation |
| `repcard_enable_send_repcard` | No | `True` | Enable sending digital business cards |
| `repcard_default_calendar` | No | (auto-synced) | Calendar ID for new appointments (dropdown populated after sync) |
| `repcard_default_setter` | Yes* | `--` | Setter (sales rep) email for new appointments |
| `repcard_default_contact_type` | No | `Lead` | Contact type for new contacts (Lead, Customer, Recruit) |
| `repcard_default_contact_owner` | Yes* | `--` | Owner (assigned rep) email for new contacts |
| `repcard_default_contact_status` | No | `New` | Status for new contacts |
| `repcard_default_repcard_id` | No | `--` | Digital business card template ID |
| `repcard_setup_scenarios` | No | `True` | Auto-add booking/cancellation scenarios to canvas |

*Required for the respective feature to work (booking requires setter, contacts require owner).

3. Click **Save + Publish All**
4. On publish, the InitFlow automatically:
    - Creates HTTP connector for Repcard API ( `repcard_connector` )
    - Sets all customer attributes with metadata
    - Registers booking, availability, and cancellation tools on ConvoAgent
    - Syncs calendars from Repcard API (GET /calendar/lists)
    - Populates the calendar dropdown with synced values

## 4. How to Use

### 4.1 How the Integration Works

The integration uses the Repcard **REST API** with `x-api-key` header authentication. All requests go through the centralized NetworkFlow, which routes to either the real HTTP connector (Production) or LLM emulation (DevMode).

- A **Trigger** is a system event that starts a flow
- An **Action** is the API operation performed in Repcard CRM

**Available Triggers**

| Trigger | Event | Description |
|---|---|---|
| **Create Appointment** | `book_slot` | When the agent confirms a booking |

| | | |
|---|---|---|
| **Cancel Appointment** | `convo_agent_cancel_booking` | When the agent processes a cancellation |
| **Check Availability** | `get_available_slots` | When the agent checks for open time slots |
| **Create Contact** | `repcard_create_contact` | When a new contact needs to be created |
| **Send RepCard** | `repcard_send_repcard` | When a digital business card is sent |
| **Setup** | `project_publish_finish` | Initialize integration on deploy |
| **Canvas Setup** | `gm_workflow_builder_canvas_built` | Auto-add scenarios to canvas |

**Available Actions**

- **Create Appointment** -- Submit new appointment (POST `/appointments`)
- **Cancel Appointment** -- Cancel appointment with notes (PATCH `/appointments/{id}`)
- **Get Appointments** -- Fetch existing appointments for availability calculation (GET `/appointments?from_date&to_date`)
- **Create Contact** -- Create new customer (POST `/customers`)
- **Send RepCard** -- Send digital business card (POST `/send-repcard`)
- **Sync Calendars** -- Fetch active calendars (GET `/calendar/lists?status=active`)

### 4.2 Flows

| Flow | Purpose |
|---|---|
| **InitFlow** | Initialize integration: connectors, attributes, calendar sync |
| **NetworkFlow** | HTTP execution layer with Production/DevMode routing |
| **ApiFlow** | Centralized request builders + result routing for all domain API calls |
| **BookingFlow** | Entry point for booking -- validates and delegates to ApiFlow |
| **CancellationFlow** | Entry point for cancellation -- validates and delegates to ApiFlow |
| **AvailabilityFlow** | Entry point for availability -- delegates to ApiFlow, computes free slots |
| **ContactFlow** | Entry point for contact creation -- validates and delegates to ApiFlow |
| **RepcardSendFlow** | Entry point for sending RepCards -- validates and delegates to ApiFlow |
| **CanvasSetupFlow** | Auto-add booking + cancellation scenarios to canvas |
| **LogFlow** | Centralized debug logging |

### 4.3 Architecture Diagrams

**Overall Sequence**

```
sequenceDiagram
    participant U as "User"
    participant A as "ConvoAgent"
    participant I as "Repcard Integration"
```

```
    participant API as "Repcard API"

    Note over I,API: "Setup (on publish)"
    A->>I: "project_publish_finish"
    I->>I: "Create connector, set attributes"
    I->>API: "GET /calendar/lists?status=active"
    API-->>I: "Calendar list"
    I->>I: "Store calendars, populate dropdown"

    Note over U,API: "Check Availability"
    U->>A: "What times are available?"
    A->>I: "get_available_slots"
    I->>API: "GET /appointments?from_date&to_date"
    API-->>I: "Existing appointments"
    I->>I: "Compute free 30-min slots"
    I->>A: "result_success {available_slots}"
    A->>U: "Here are the available times..."

    Note over U,API: "Booking"
    U->>A: "Book me for Tuesday at 10am"
    A->>A: "Collect: name, phone, email, address, date/time"
    A->>I: "book_slot {customerInfo, bookingDateTime, calendarId}"
    I->>API: "POST /appointments"
    API-->>I: "{id, status}"
    I->>A: "result_success {booking_id}"
    A->>U: "Your appointment is confirmed!"

    Note over U,API: "Cancellation"
    U->>A: "Cancel my appointment"
    A->>I: "convo_agent_cancel_booking {booking_id}"
    I->>API: "PATCH /appointments/{id}"
    API-->>I: "success"
    I->>A: "result_success"
    A->>U: "Your appointment has been cancelled"
```

**InitFlow**

```
flowchart TD
    E["project_publish_finish"] --> CHECK{"target_reference
==\nrepcard_integration?"}
    CHECK -->|"No"| SKIP["Return"]
    CHECK -->|"Yes"| CALS_RAW["Load repcard_calendars\nfrom attributes"]
    CALS_RAW --> BUILD_ENUM["Build calendar enum\nfor booking schema"]
    BUILD_ENUM --> SET_SA["Set superagent attributes:\nbooking,
availability,\ncancellation enabled"]
    SET_SA --> SET_SCHEMA["Set booking payload\nschema with calendarId enum"]
    SET_SCHEMA --> SET_PA["Set project attributes:\nmode, api_key, base_url,\nenable
flags, defaults"]
    SET_PA --> CONNECTOR["Create repcard_connector\n(HTTP integration)"]
    CONNECTOR --> PERSONA["Create SetupPersona\nfor background tasks"]
    PERSONA --> KEY_CHECK{"repcard_api_key\nconfigured?"}
```

```
    KEY_CHECK -->|"No"| DONE["Done"]
    KEY_CHECK -->|"Yes"| SYNC["Send repcard_sync_calendars_request"]
    SYNC --> DONE
```

**NetworkFlow -- Production vs DevMode**

```
flowchart TD
    REQ["repcard_execute_request"] --> MODE{"repcard_mode?"}
    MODE -->|"Production"| REAL["_executeRequestSkill\nBuild URL, set x-api-key
header"]
    MODE -->|"DevMode"| EMULATE["_emulateRequestSkill\nGen() LLM
generates\nrealistic response"]
    REAL --> CONNECTOR["SendCommand to\nrepcard_connector"]
    CONNECTOR --> RESP_OK["result_success\nfrom HTTP connector"]
    CONNECTOR --> RESP_ERR["result_error\nfrom HTTP connector"]
    RESP_OK --> STATUS{"statusCode\n200 or 201?"}
    STATUS -->|"Yes"| SUCCESS["repcard_result_success"]
    STATUS -->|"No"| ERROR["repcard_result_error"]
    RESP_ERR --> ERROR
    EMULATE --> SUCCESS
```

**ApiFlow**

```
flowchart TD
    subgraph "Request Builders"
        CA["CreateAppointmentApiSkill\nPOST /appointments"]
        CC["CreateContactApiSkill\nPOST /customers"]
        SR["SendRepcardApiSkill\nPOST /send-repcard"]
        CANCEL["CancelAppointmentApiSkill\nPATCH /appointments/{id}"]
        AVAIL["GetAvailableSlotsApiSkill\nGET /appointments"]
        SYNC["SyncCalendarsApiSkill\nGET /calendar/lists"]
    end

    subgraph "Result Router"
        ERS["ExecuteResultSuccessSkill\nRoute by runSkill"]
        ERE["ExecuteResultErrorSkill\nRoute by errorSkill"]
    end

    subgraph "Success Handlers"
        CAS["_createAppointmentSuccessSkill\nemit repcard_booking_success"]
        CCS["_createContactSuccessSkill\nemit repcard_contact_success"]
        SRS["_sendRepcardSuccessSkill\nemit repcard_send_repcard_success"]
        CANS["_cancelAppointmentSuccessSkill\nemit repcard_cancellation_success"]
        BAS["_buildAvailableSlotsSkill\ncompute free slots,\nemit
repcard_availability_success"]
        SCS["_syncCalendarsSuccessSkill\nstore calendars,\nupdate dropdown"]
    end

    CA --> |"repcard_execute_request"| ERS
    CC --> |"repcard_execute_request"| ERS
```

```
    SR --> |"repcard_execute_request"| ERS
    CANCEL --> |"repcard_execute_request"| ERS
    AVAIL --> |"repcard_execute_request"| ERS
    SYNC --> |"repcard_execute_request"| ERS

    ERS --> |"runSkill routing"| CAS
    ERS --> |"runSkill routing"| CCS
    ERS --> |"runSkill routing"| SRS
    ERS --> |"runSkill routing"| CANS
    ERS --> |"runSkill routing"| BAS
    ERS --> |"runSkill routing"| SCS

    ERE --> |"errorSkill routing"| ERR_HANDLERS["Error emit skills\n(per domain)"]
```

**BookingFlow**

```
flowchart TD
    E["book_slot"] --> GATE{"repcard_enable_booking\n== True?"}
    GATE -->|"No"| SKIP["Return"]
    GATE -->|"Yes"| DELEGATE["Send repcard_book_slot_request\nto ApiFlow"]
    DELEGATE --> API["CreateAppointmentApiSkill"]
    API --> VALIDATE{"api_key + firstName +\nphone + bookingDateTime +\ncalendarId +
setter?"}
    VALIDATE -->|"No"| ERR["repcard_booking_error"]
    VALIDATE -->|"Yes"| BUILD["Build POST /appointments body:\ncalendarId,
setterEmail,\nstartAt, endAt, timezone,\ncustomer {name, phone, email, address}"]
    BUILD --> EXEC["repcard_execute_request"]
    EXEC --> OK{"Success?"}
    OK -->|"No"| ERR
    OK -->|"Yes"| EXTRACT["Extract appointment ID\nfrom response"]
    EXTRACT --> SUCCESS["repcard_booking_success\n{booking_id, bookingDateTime}"]
    SUCCESS --> RESULT["result_success\ntargetAction: book_slot"]
```

**CancellationFlow**

```
flowchart TD
    E["convo_agent_cancel_booking"] --> GATE{"repcard_enable_cancellation\n==
True?"}
    GATE -->|"No"| SKIP["Return"]
    GATE -->|"Yes"| DELEGATE["Send repcard_cancel_booking_request\nto ApiFlow"]
    DELEGATE --> API["CancelAppointmentApiSkill"]
    API --> VALIDATE{"api_key +\nbooking_id?"}
    VALIDATE -->|"No"| ERR["repcard_cancellation_error"]
    VALIDATE -->|"Yes"| BUILD["Build PATCH /appointments/{id}\nbody: {notes}"]
    BUILD --> EXEC["repcard_execute_request"]
    EXEC --> OK{"Success?"}
    OK -->|"No"| ERR
    OK -->|"Yes"| SUCCESS["repcard_cancellation_success\n{booking_id}"]
    SUCCESS --> RESULT["result_success\ntargetAction: cancel_booking"]
```

**AvailabilityFlow**

```
flowchart TD
    E["get_available_slots"] --> DELEGATE["Send
repcard_check_availability_request\nto ApiFlow"]
    DELEGATE --> API["GetAvailableSlotsApiSkill"]
    API --> VALIDATE{"api_key\nconfigured?"}
    VALIDATE -->|"No"| ERR["repcard_availability_error"]
    VALIDATE -->|"Yes"| BUILD["Build GET /appointments\nquery:
from_date=today,\nto_date=today+5, per_page=100"]
    BUILD --> EXEC["repcard_execute_request"]
    EXEC --> OK{"Success?"}
    OK -->|"No"| ERR
    OK -->|"Yes"| SLOTS["_buildAvailableSlotsSkill:\nParse busy times from
appointments\nGenerate 30-min slots (9am-5pm)\nExclude busy + past slots\nReturn top
20"]
    SLOTS --> SUCCESS["repcard_availability_success\n{available_slots,
total_count}"]
    SUCCESS --> RESULT["result_success\ntargetAction: get_available_slots"]
```

**ContactFlow**

```
flowchart TD
    E["repcard_create_contact"] --> GATE{"repcard_enable_contact_creation\n==
True?"}
    GATE -->|"No"| SKIP["Return"]
    GATE -->|"Yes"| DELEGATE["Send repcard_create_contact_request\nto ApiFlow"]
    DELEGATE --> API["CreateContactApiSkill"]
    API --> VALIDATE{"api_key + firstName +\nphone + owner?"}
    VALIDATE -->|"No"| ERR["repcard_contact_error"]
    VALIDATE -->|"Yes"| BUILD["Build POST /customers body:\nfirstName, lastName,
userEmail,\ntype, statusName, email,\nphoneNumber, contactSource=AI"]
    BUILD --> EXEC["repcard_execute_request"]
    EXEC --> OK{"Success?"}
    OK -->|"No"| ERR
    OK -->|"Yes"| SUCCESS["repcard_contact_success"]
```

**RepcardSendFlow**

```
flowchart TD
    E["repcard_send_repcard"] --> GATE{"repcard_enable_send_repcard\n== True?"}
    GATE -->|"No"| SKIP["Return"]
    GATE -->|"Yes"| DELEGATE["Send repcard_send_repcard_request\nto ApiFlow"]
    DELEGATE --> API["SendRepcardApiSkill"]
    API --> VALIDATE{"api_key + contact_id +\nrepcard_id?"}
    VALIDATE -->|"No"| ERR["repcard_send_repcard_error"]
    VALIDATE -->|"Yes"| BUILD["Build POST /send-repcard body:\nexisting_contact,
sms_or_email,\nrepcard, message"]
    BUILD --> EXEC["repcard_execute_request"]
```

```
    EXEC --> OK{"Success?"}
    OK -->|"No"| ERR
    OK -->|"Yes"| SUCCESS["repcard_send_repcard_success"]
```

**CanvasSetupFlow**

```
flowchart TD
    E["gm_workflow_builder_canvas_built"] --> CHECK{"repcard_setup_scenarios\n==
False?"}
    CHECK -->|"Yes"| SKIP["Return"]
    CHECK -->|"No"| LIB["SetupLibrarySkill"]
    LIB --> ITEMS["Define items to insert:\n- cancellation_via_agent_intent\n-
library_intent_types_common_appointment_agent\n- cancellation_via_agent_scenario\n-
repcard_schedule_appointment_via_agent_scenario"]
    ITEMS --> LOOP["For each item:\nCheck if already on canvas\nIf not, get from
_getScenariosSkill\nor library"]
    LOOP --> DELETE["Delete old generic
items:\nscenario_home_service_appointment_request\nintent_type_appointment_scheduling\n
    DELETE --> ADD["Add Repcard-specific items\nto canvas via batch"]
```

**LogFlow**

```
flowchart TD
    E["repcard_log"] --> LOG["LogSkill:\nDUMMY(Repcard: message)"]
```

### 4.4 Canvas Scenarios

When `repcard_setup_scenarios` is `True` (default), the CanvasSetupFlow automatically adds:

**Scenarios:**

- **"Scheduling Appointment via Agent"** ( `repcard_schedule_appointment_via_agent_scenario` ) -- 10-step guided booking flow that collects all required fields (name, phone, email, address, ZIP, date/time) with code-phrases
- **"Canceling Appointment via Agent"** ( `cancellation_via_agent_scenario` ) -- 6-step cancellation flow with confirmation and code-phrases

**Intent Types:**

- **"[L] Appointment via Agent"** ( `library_intent_types_common_appointment_agent` ) -- from library
- **"[T] Cancellation via Agent"** ( `cancellation_via_agent_intent` ) -- caller wants to cancel

### 4.5 Where to Test

Testing can be done through the **Newo conversation interface** (chat or voice).

**DevMode:** Set `repcard_mode` to `DevMode` to test without real API credentials. The integration will use LLM to generate realistic Repcard API responses.

### 4.6 Testing and Verification

**To test the integration:**

1. **Setup**: Publish the project and verify calendar sync completed (check `repcard_calendars` attribute and `repcard_default_calendar` dropdown)
2. **Availability**: Ask about available times -- verify agent returns computed free slots
3. **Booking**: Request an appointment -- provide name, phone, email, address, and date --> verify appointment created in Repcard
4. **Cancellation**: Ask to cancel -- verify appointment status updated in Repcard
5. **Contact Creation**: Trigger contact creation -- verify customer created in Repcard via POST /customers

**If no action occurs:**

- Ensure API key is set ( `repcard_api_key` ) -- requires Enterprise account
- Verify `repcard_mode` is `Production` (not `DevMode` )
- Check feature flags ( `repcard_enable_booking` , `repcard_enable_cancellation` , etc.)
- Confirm `repcard_default_setter` is set (required for booking)
- Confirm `repcard_default_contact_owner` is set (required for contacts)
- Verify calendars synced (check `repcard_default_calendar` dropdown)
- Re-publish after configuration changes

> *Note: This integration creates real appointments and contacts in Repcard CRM. Changes made through the agent are reflected in the live system.*

## 5. FAQ

**Q: How does authentication work?** A: Repcard uses an **API Key** sent in the `x-api-key` HTTP header. API access requires an Enterprise-tier Repcard CRM account. The key is obtained from Company Settings > API Keys tab.

**Q: What is DevMode?** A: When `repcard_mode` is set to `DevMode` , the integration uses LLM (Gen) to emulate Repcard API responses instead of making real HTTP requests. Useful for testing the booking flow without creating real appointments.

**Q: How does calendar selection work?** A: On publish, the integration calls `GET /calendar/lists?status=active` to fetch all active calendars. These are stored in `repcard_calendars` and populate the `repcard_default_calendar` dropdown. The LLM can also select a `calendarId` from the enum in the booking schema. If no calendar is selected, the default is used.

**Q: Why is the customers endpoint POST /customers and not POST /contacts?** A: Repcard CRM uses `/customers` as the endpoint name for creating contacts. This is the official Repcard API convention despite the feature being called "contact creation" in the integration.

**Q: How does availability checking work?** A: The integration fetches all existing appointments for the next 5 days via `GET /appointments?from_date&to_date` , then computes free 30-minute slots during business hours (9:00 AM to 5:00 PM) by excluding busy time slots. Past slots are also excluded. Up to 20 available slots are returned.

**Q: What DateTime format does the API use?** A: The Repcard API uses `YYYY-MM-DD HH:MM:SS` format. The integration automatically appends `:00` seconds if the input is in `YYYY-MM-DD HH:MM` format (16 characters).

**Q: How does the calendarId work?** A: The `calendarId` is numeric and identifies a calendar in Repcard CRM. It is synced dynamically from the API. The `setterEmail` identifies the sales representative assigned

to the appointment.

**Q: Can I use the Send RepCard feature through the conversational agent?** A: The Send RepCard feature ( `POST /send-repcard` ) is primarily designed for Zapier-style automation triggers. It requires a `contact_id` and a `repcard_default_repcard_id` template to be configured.